AHMAD RAMAZANLI

Machine Learning Engineer

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SUMMARY

Machine learning engineer with a software development background, specializing in time-series forecasting and predictive modeling for the energy sector. Currently developing an LSTM-based model in PyTorch for solar energy production forecasting, with hands-on experience in regression, classification, and statistical modeling across Python, R, and MATLAB. Author of a peer-reviewed publication in the electric power domain.

WORK EXPERIENCE

Chief Specialist — Project and Research Institute, Azerenergy

Baku, Azerbaijan | 11/2022 – Present

- Develop computational and soft-computing models (MATLAB) for assessment and decision-support tasks in the national energy system.
- Apply machine learning and statistical methods to energy-sector data; co-authored a peer-reviewed journal publication.

Data Science Intern (Mentor) — QSS Analytics

04/2022 – 08/2022

- Delivered machine learning and data analysis projects across several domains under supervision.
- Mentored students through lessons and hands-on ML project work.

Software Developer — Hacettepe University, Chemistry Department

Ankara, Turkey | 08/2017 – 06/2018

- Developed C++ plugins for the Psi4 quantum chemistry package, implementing matrix algebra models for MP2 and CIS methods.

SKILLS

Machine Learning: PyTorch, deep learning (LSTM), time-series forecasting, regression models, classification (logistic regression), PCA, model evaluation

Data: NumPy, pandas, statistical analysis, data cleaning, feature engineering

Programming: Python, R, MATLAB, C++, ARM Assembly

Tools: Git, GitHub, Excel

PROJECTS

Solar Energy Production Forecasting — Ongoing personal research project

- Building an LSTM-based deep learning model in PyTorch to forecast solar panel energy production from historical time-series data.
- Iterating on earlier machine learning and regression approaches to improve forecast accuracy.

Short-Circuit Prediction in Multi-Phase Electrical Systems — Python

- Developed a machine learning model to predict short circuits in multi-phase electrical systems.

Customer Churn Analysis — R

- Built a logistic regression model to predict customer churn.

Medical Data Analysis — Python

- Applied Principal Component Analysis for dimensionality reduction and pattern discovery in medical data.

Quantum Chemistry Plugins for Psi4 — C++

- Implemented density-fitted MP2 (Møller–Plesset perturbation theory) and Configuration Interaction Singles (CIS) modules; integrated an existing Davidson eigensolver to converge excited states.

PUBLICATIONS

Nasibov V.Kh., Alizade R.R., Mastaliyev I.Y., Ramazanli A.M. “Application of the Fuzzy-Set Theory to Assess the Knowledge of Electric Power Industry Specialists.” Reliability: Theory & Applications, Vol. 19, No. 1 (77), p. 424, March 2024.

Ramazanli A. “A Unified Framework for Divisibility Rules of Prime Numbers.” Preprint, Zenodo, April 2026. DOI: 10.5281/zenodo.19925110